

Dhinesh Karthick D

Electronics & Communication Engineering Student | VLSI Design | Embedded Systems |

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SUMMARY

Electronics and Communication Engineering student specializing in VLSI design and embedded systems, with hands-on experience in analog and digital IC design, ultra-low-power circuit design, and edge-AI sensor fusion. Builds full-stack embedded and AI-driven engineering solutions across hardware and software.

EDUCATION

Chennai Institute of Technology

2025 – 2029

Bachelor of Engineering, Electronics & Communication Engineering CGPA – 9.03

Kalashetra Matric Hr Sec School

Higher Secondary Education +2 - 96.3 %

Secondary School Leaving Certificate - 95.6 %

EXPERIENCE

VLSI Design Intern — Centre for Advanced Chip Design (Centre of Excellence)

2026

- Performed analog and digital IC design in Cadence Virtuoso, covering schematic design, custom layout, and simulation.
- Designed an ultra-low-power flip-flop as a capstone project, including full DRC/LVS verification and post-layout simulation.

PROJECTS

[AegisLink](#) — Two-Wheeler Crash Detection System

2026

- Built an ESP32 + MPU6050-based crash detection system with BLE telemetry to a Flutter companion app, designed for zero-internet-dependency operation.
- Diagnosed and resolved a BLE protocol mismatch between firmware and app; replaced ML-based classification with a calibrated threshold detection approach after model availability constraints.

[EdgeWatt](#) — Non-Intrusive Load Monitoring (NILM) System

2026

- Developed an ESP32 + CT sensor-based system to classify appliance loads from a single monitoring point using steady-state RMS and crest-factor features.
- Built a 3-class nearest-neighbor classifier calibrated on real measurements, with live power/energy estimation and an HTML dashboard.

[GeoDrain AI](#) — AI Powered Geospatial Decision Support System

2026

- GeoDrain AI is an AI-assisted geospatial platform designed to reduce the manual effort involved in drainage planning for rural and semi-urban local bodies.
- The system combines terrain intelligence, hydrological analysis, geospatial feature extraction, drainage optimization, and hydraulic model preparation into a unified workflow.

TECHNICAL SKILLS

- Languages: C, C++, Python, Dart
- VLSI / EDA: Cadence Virtuoso, Schematic Design, Custom Layout, DRC/LVS Verification, Verilog
- Embedded Systems: ESP32, Arduino, PCB Design, Sensor Integration, IoT Systems
- AI / ML: Machine Learning fundamentals, Edge Impulse, applied ML pipelines
- Tools: Git, GitHub, LTSpice, Flutter, Linux, VS Code

ACHIEVEMENTS

- IITM I2I Challenge — Top 50 (IIT Madras)
- H@CIT Codeathon — Finalist
- Geonex Hackathon, CEG Anna University — Finalist
- APATHON — Top 100
- 200+ problems solved on LeetCode